

Question Paper

Exam Date & Time: 25-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Human Computer Interaction [NPT_4131]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Apply the principles of User-Centered Design (UCD) to an IoT-based home automation system. (5)
Experiment with different interaction modalities (touch, voice, gesture) and explain how they can solve usability issues for elderly users.
- A)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Analyze the differences between Graphical User Interface (GUI) and Natural User Interface (NUI). (3)
Compare their usability in real-world contexts and provide a short conclusion about which is more effective for accessibility.
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- C) Inspect how ethical considerations influence the IRB approval process and discover one possible risk if these standards are not followed. (2)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 2) Utilize the concept of *triangulation* to describe how multiple data sources can be combined to enhance the reliability of findings in HCI research. (5)
- A)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Solve the issue of *data misinterpretation* by proposing three steps that ensure accurate statistical analysis and reporting in quantitative research. (3)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- C) Identify two common pitfalls that researchers often face while applying for IRB approval and briefly explain how these can impact the research timeline. (2)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 3) Apply the concept of the Double Diamond design model to explain how prototyping fits within the stages of *divergent and convergent thinking*. Illustrate your answer with an example of designing an accessible user interface for visually impaired users. (5)
- A)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Given a Mid-Fi student-portal prototype, examine the design and inspect its core flows. Test for three usability/accessibility risks specifically highlighted in the talk (e.g., navigation clarity,

affordance indicators like dropdown arrows, font/contrast scalability).

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- C) Demonstrate how two ethical challenges researchers may arise during data gathering can be minimized through IRB-approved protocols. (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 4) Compare the role of Lo-Fi and Hi-Fi prototyping in refining a product's design. Contrast how each stage contributes differently to user feedback and design iteration. (5)

A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Apply three key factors to decide when to transition from low-fidelity to mid-fidelity prototypes in an iterative design process. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Divide the IoT design challenges mentioned into *technical* and *ethical* categories, and discover one key user-centered solution for each. (2)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Apply your understanding of qualitative and quantitative data gathering methods to design an experiment that measures user satisfaction with a newly developed interface. Explain how you would balance statistical validity with ethical considerations. (5)

A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Categorize the different data gathering methods used in HCI research. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) Utilize a conversational AI tool (such as Alexa or ChatGPT) to explain how HCI design principles like accessibility and user engagement can improve user interaction quality. (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 NPT_4131:Human Computer Interaction

#	Course Outcome	Description	Marks
1	1	CO 1	7
2	2	CO 2	5
3	3	CO 3	23
4	4	CO 4	15
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 NPT_4131:Human Computer Interaction

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)

3	Application	9	32(64%)
4	Analysis	6	18(36%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)